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| <b>SHALEEN JAYARAJ</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                            | <a href="https://www.linkedin.com/in/shaleen-jayaraj-267859167/">linkedin.com/in/shaleen-jayaraj-267859167/</a> |
| <b>Vehicle Dynamics CAE Engineer   M.S. Automotive Engineering</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                            |                                                                                                                 |
| Graduate Automotive Engineering student at Clemson University (CU-ICAR) with nearly 3 years of professional experience in vehicle dynamics simulation and multibody dynamics (MBD). Experienced in ride & handling analysis, K&C studies, vehicle stability control, and CAE model correlation. Proficient in MATLAB/Simulink, CarSim, and MSC ADAMS, with experience developing high-fidelity vehicle models and automating CAE workflows.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                            |                                                                                                                 |
| <b>Greenville, South Carolina</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>shaleej@clemson.edu</b> | <b>+1 864-451-8757</b>                                                                                          |
| <b>Skills</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                            |                                                                                                                 |
| <b>Controls &amp; Dynamics:</b> Vehicle Dynamics, Ride & Handling, Kinematics & Compliance (K&C), Multibody Dynamics (MBD), Vehicle Stability Control (VSC/ESC), Nonlinear Control (FBL, SMC), Model Predictive Control (MPC), State Estimation<br><b>Simulation &amp; Tools:</b> MATLAB/Simulink, CarSim, MSC ADAMS Car, ADAMS View, Siemens Test Lab, Altair Hypermesh<br><b>Programming:</b> Python, MATLAB, ADAMS CMD Scripting<br><b>CAD:</b> SolidWorks, Siemens NX                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                            |                                                                                                                 |
| <b>Professional Experience</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                            |                                                                                                                 |
| <b>Vehicle Dynamics Engineer - Applus+ IDIADA (TATA Motors ODC)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                            | <b>Pune, India   10th Aug'22 – 25th Apr'25</b>                                                                  |
| <ul style="list-style-type: none"> <li>Led transient and steady-state vehicle dynamics analyses (cornering, braking, step steer, ride) to evaluate and improve handling and stability for passenger vehicles.</li> <li>Modified, maintained, and upgraded existing multi-body dynamics (MBD) models in ADAMS/View; ensured model integrity, version control, and traceability for ongoing development programs.</li> <li>Developed and documented team processes, scripts, and automation tools that were integral to CAE workflow efficiency, model consistency, and cross-team collaboration.</li> <li>Created simplified vehicle and suspension dynamics models based on first principles to support early concept studies, design trade-offs, and engineering decision-making.</li> <li>Built full-vehicle multibody models using Template Builder, Event Builder, and Road Builder, including flex-body integration for suspension and chassis components.</li> <li>Created vehicle settling procedures in simulation environments to establish accurate static equilibrium conditions prior to dynamic events.</li> <li>Supported engineering teams by aiding modeling studies, simulation correlation, and performance investigations across multiple vehicle programs.</li> <li>Executed Kinematics &amp; Compliance (K&amp;C) studies to quantify bump steer, roll steer, camber compliance, and wheel rates; proposed geometry and bushing changes to improve handling predictability and steering response.</li> <li>Applied Design of Experiments (DOE) in ADAMS/Insight to optimize suspension hardpoints and meet ride and handling targets across varying load conditions.</li> <li>Diagnosed steering wobble and rack-load issues; optimized hydrobush stiffness and steering link geometry, achieving ~\$25 USD/vehicle cost savings.</li> <li>Performed geometric and durability checks including Driveshaft pull out, RLDA, wheel/ARB envelopes, service load validation, and ground-clearance audits.</li> <li>Collaborated cross-functionally with chassis, steering, and CAE teams to validate results and deliver correlated simulation-to-test reports.</li> </ul> |                            |                                                                                                                 |
| <b>Education</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                            |                                                                                                                 |
| <b>M.S., Automotive Engineering — Clemson University CUICAR 2025 – 2027   GPA: 3.71/4.0</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                            |                                                                                                                 |
| Coursework: Dynamic Performance of Vehicles, Control Systems Engineering, Vehicle Testing & Characterization, Automotive Safety & Stability, Tires, Automotive Light weight optimization                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                            |                                                                                                                 |
| <b>B.Tech. (Hons.), Mechanical Engineering — Rajiv Gandhi Institute of Technology, APJAKTU 2018 – 2022   GPA: 8.32/10</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                            |                                                                                                                 |
| <b>Project Work (AUE Clemson):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                            |                                                                                                                 |
| <b>Nonlinear Vehicle Stability Control (Simulink and CarSim)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                            |                                                                                                                 |
| <ul style="list-style-type: none"> <li>Developed Feedback Linearization (FBL) and Sliding Mode Control (SMC) algorithms for yaw-rate tracking under FMVSS 126 Sine-with-Dwell and ISO 3888-1 Double Lane Change maneuvers.</li> <li>Built and validated a 7-DOF nonlinear vehicle dynamics model in MATLAB/Simulink against high-fidelity CarSim reference models.</li> <li>Implemented nonlinear tire-force and state-dependent cornering stiffness models to capture realistic vehicle handling and tire saturation behavior.</li> <li>Evaluated controller performance through yaw-rate tracking, lateral acceleration response, trajectory stability, and control effort under varying speeds and maneuver conditions.</li> <li>Compared FBL and SMC strategies, demonstrating improved vehicle stability and reduced control chattering through robust sliding mode control design.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                            |                                                                                                                 |
| <b>Vehicle Dynamics Data Analysis &amp; Tire Model-Based Simulation</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                            |                                                                                                                 |
| <ul style="list-style-type: none"> <li>Analyzed on-track vehicle dynamics data using MATLAB to extract tire parameters, quantify slip angles, steering response delay, and evaluate lateral tire grip utilization during aggressive maneuvers.</li> <li>Estimated front and rear axle cornering stiffness using nonlinear least-squares curve fitting with velocity-based binning to capture load-transfer-dependent tire behavior.</li> <li>Developed and validated a nonlinear bicycle vehicle dynamics model in MATLAB by integrating fitted tire force characteristics and simulating fishhook and sine-sweep steering maneuvers.</li> <li>Simulated and evaluated yaw rate, lateral acceleration, and overall vehicle stability, validating model predictions against experimental test data.</li> <li>Performed vehicle instrumentation, DAQ integration, and post-processing using MATLAB and Siemens Testlab to extract frequency response functions, ride comfort metrics, vibration modes, and dynamic characteristics.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                            |                                                                                                                 |
| <b>Ride and Handling Analysis of 2-Axle and 3-Axle Trucks Using Bicycle and Ride Models.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                            |                                                                                                                 |
| <ul style="list-style-type: none"> <li>Used PSD and RMS acceleration metrics to evaluate ride comfort of 2-axle and 3-axle trucks.</li> <li>Built bicycle models and performed transient handling analysis, comparing understeer gradient, yaw response, and stability characteristics.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                            |                                                                                                                 |
| <b>Lightweight Skateboard Optimization</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                            |                                                                                                                 |
| <ul style="list-style-type: none"> <li>Utilized HyperMesh to perform topology optimization, reducing overall weight while maintaining structural integrity and performance requirements.</li> <li>Designed and 3D printed a lightweight skateboard under dimensional and mid-span deflection constraints (20 mm maximum), achieving the lowest overall weight among all designs.</li> <li>Conducted structural analysis and incorporated manufacturability considerations to develop an efficient and practical skateboard design.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                            |                                                                                                                 |
| <b>Leadership Experience</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                            |                                                                                                                 |
| <b>Vehicle Dynamics Head - Team Motorage BAJA SAE INDIA</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                            |                                                                                                                 |
| <ul style="list-style-type: none"> <li>Led the design of steering and suspension geometry using SolidWorks, followed by kinematic analysis using Lotus Shark.</li> <li>Conducted structural analysis of suspension components using ANSYS to ensure durability and performance.</li> <li>Modeled the vehicle in IPG Carmaker and performed virtual simulations to optimize vehicle dynamics and handling performance.</li> <li>Coordinated with other subsystem teams to ensure seamless integration of the suspension system with chassis and drivetrain.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                            |                                                                                                                 |
| <b>Certifications &amp; Skills</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                            |                                                                                                                 |
| <b>MSC ADAMS Certifications:</b> ADM740 – Vehicle Modeling using ADAMS/Car ADM704B – ADAMS/View Scripting, Macros & GUI Customization<br>ADM705 – Python Scripting in ADAMS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                            |                                                                                                                 |